

Data Visualization and Storytelling

Turning Data into Insightful Decisions



Presented by Manu Sharan Kumar

Data Visualization

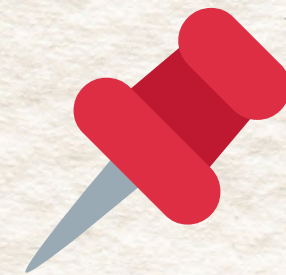
Data visualization is the graphical representation of information and data using visual elements like charts, graphs, maps, and dashboards.



“Humans process visual information far more efficiently than raw numbers or text.”

Why Data Visualization Matters ?

- Humans process visuals 60,000× faster than text
- 90% of information transmitted to brain is visual
- Raw data → Hard to understand
- Visualized data → Easy decisions



**“Visualization
transforms data
into
understanding.”**

Module Overview

**Principles
of
Effective
Visualization**

**Matplotlib
Visualizations**

**Seaborn
Statistical
Plots**

**Interactive
Visualization with
Plotly**

**Insight
Generation**

**Case Study
Project**



Principles of Effective Visualization & Dashboard Design

What Makes a Good Visualization?

Clarity

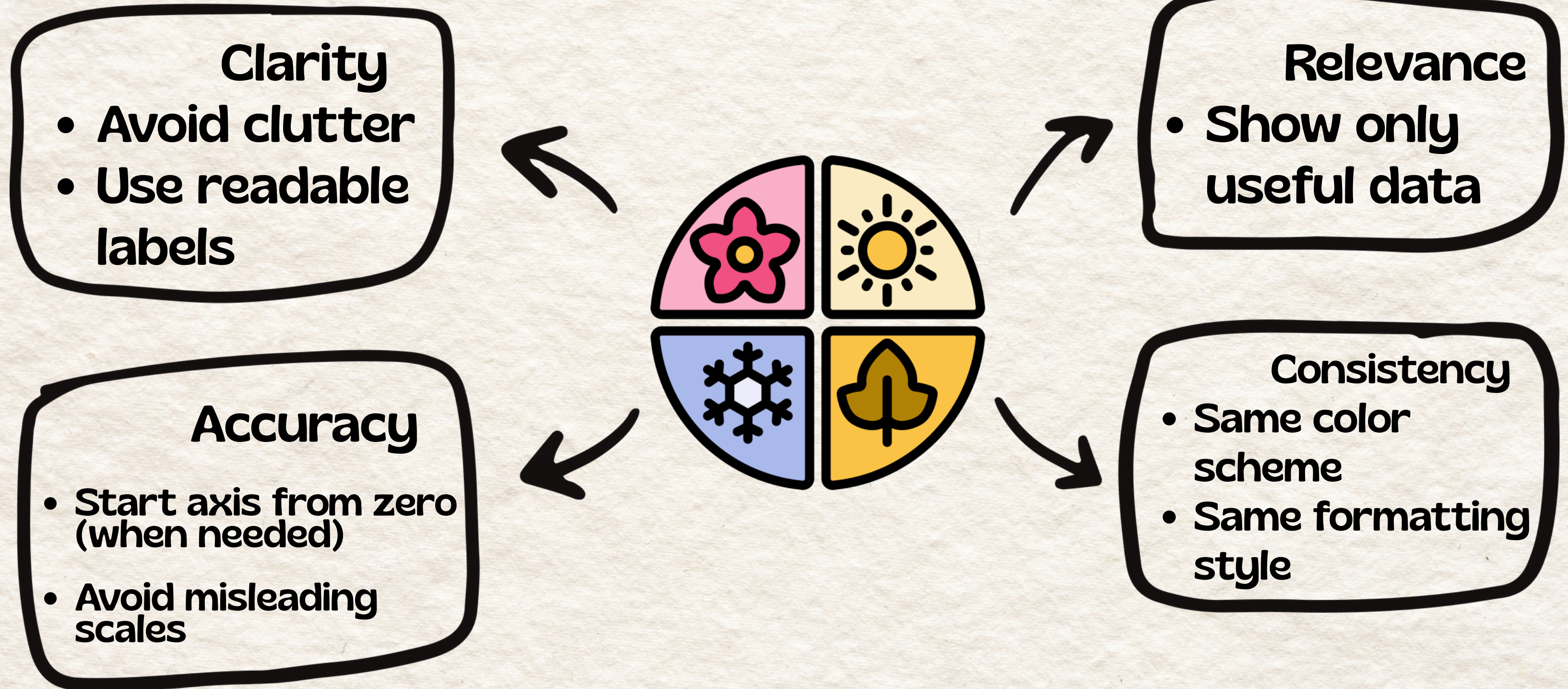
Simplicity

**Story
Telling**

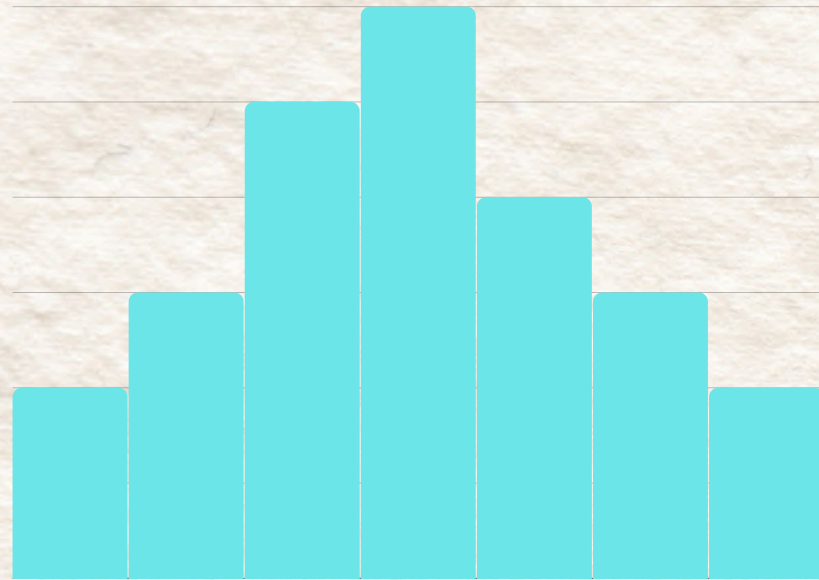
**Insight
Focused**



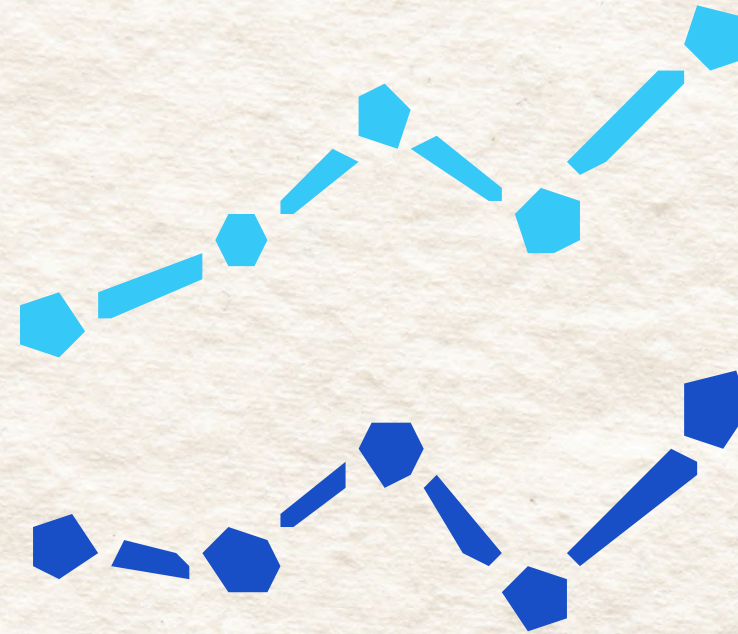
Key Principles



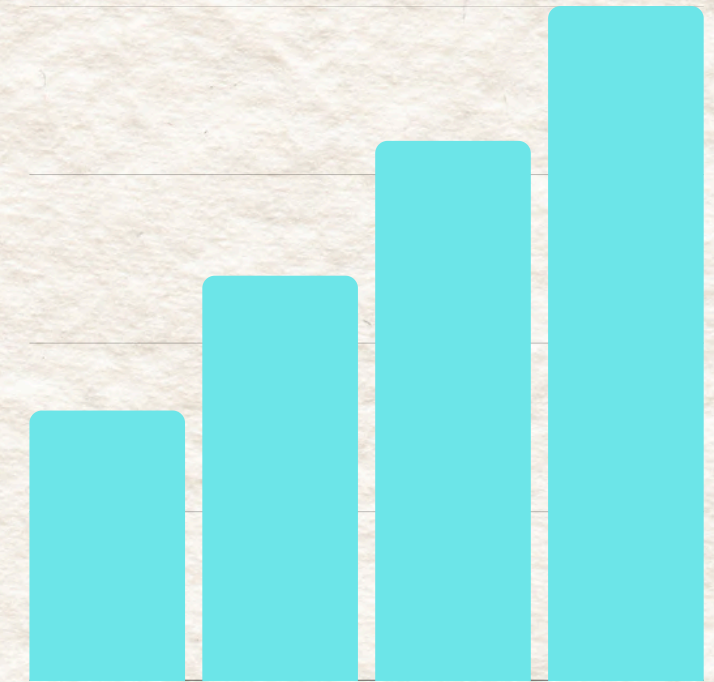
Visualization Charts



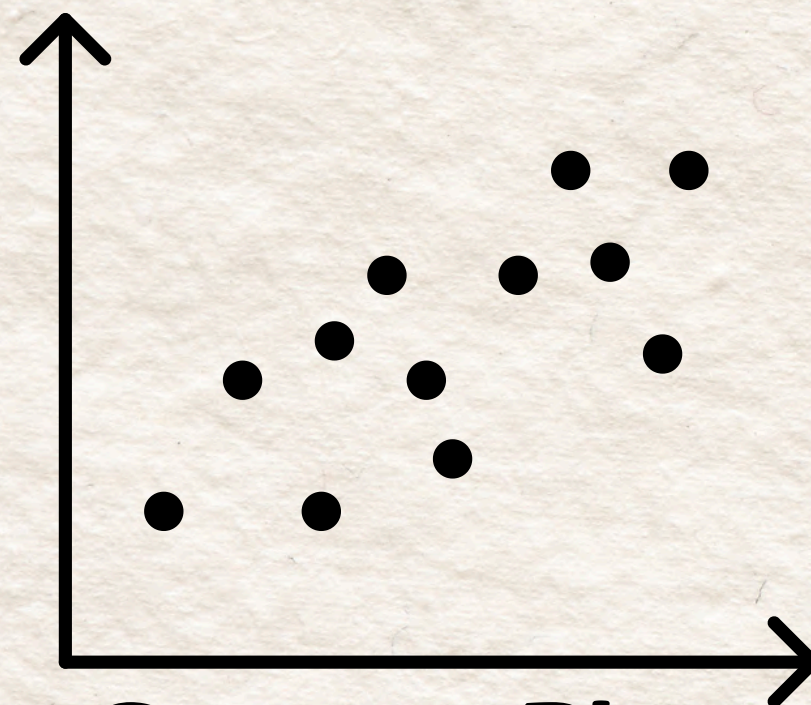
Histogram



Line Chart



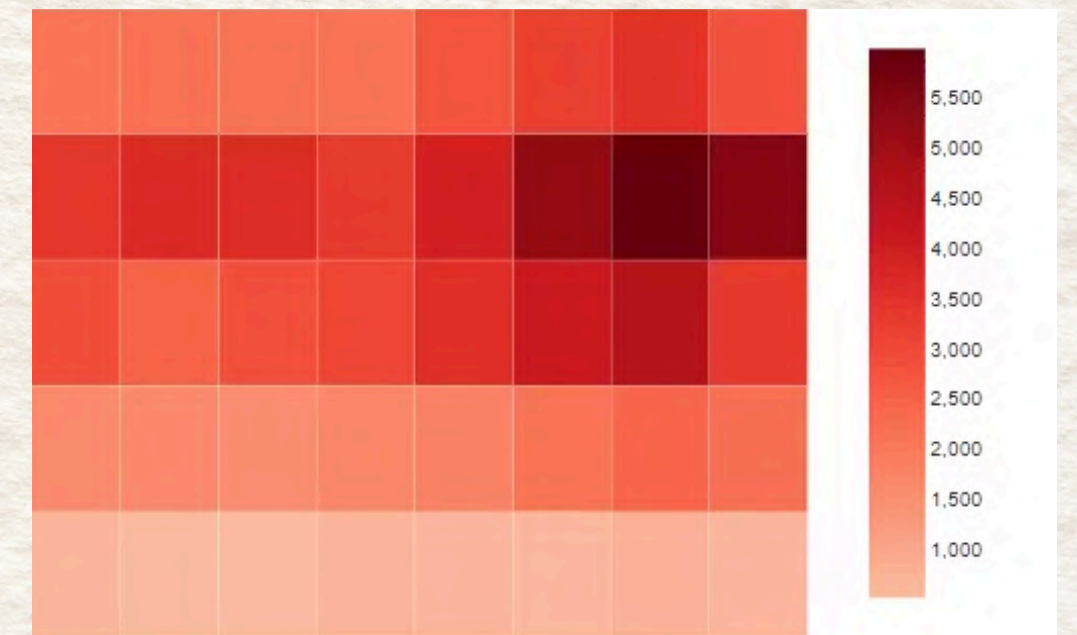
Bar Chart



Scatter Plot



Pie Chart



Heatmap

Choosing the Right Chart

Purpose	Chart Type
Trend over time	Line chart
Compare categories	Bar chart
Distribution	Histogram
Relationship	Scatter plot
Correlation	Heatmap

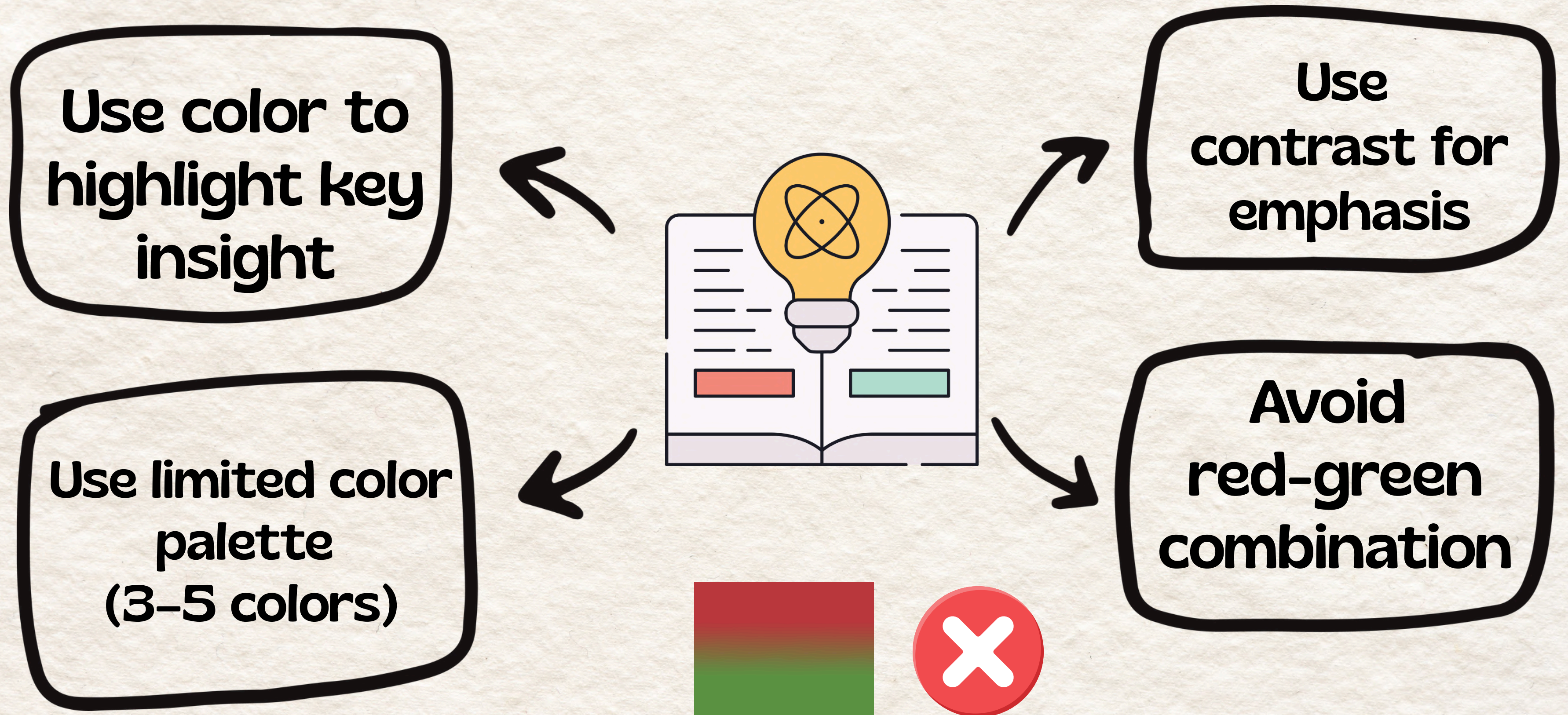
Dashboard Design Principles



**A good
dashboard
has**

- **One clear purpose**
- **Logical Layout**
- **Minimal text**
- **Consistent colors**

Color Theory in Visualization



**Too many
colors**

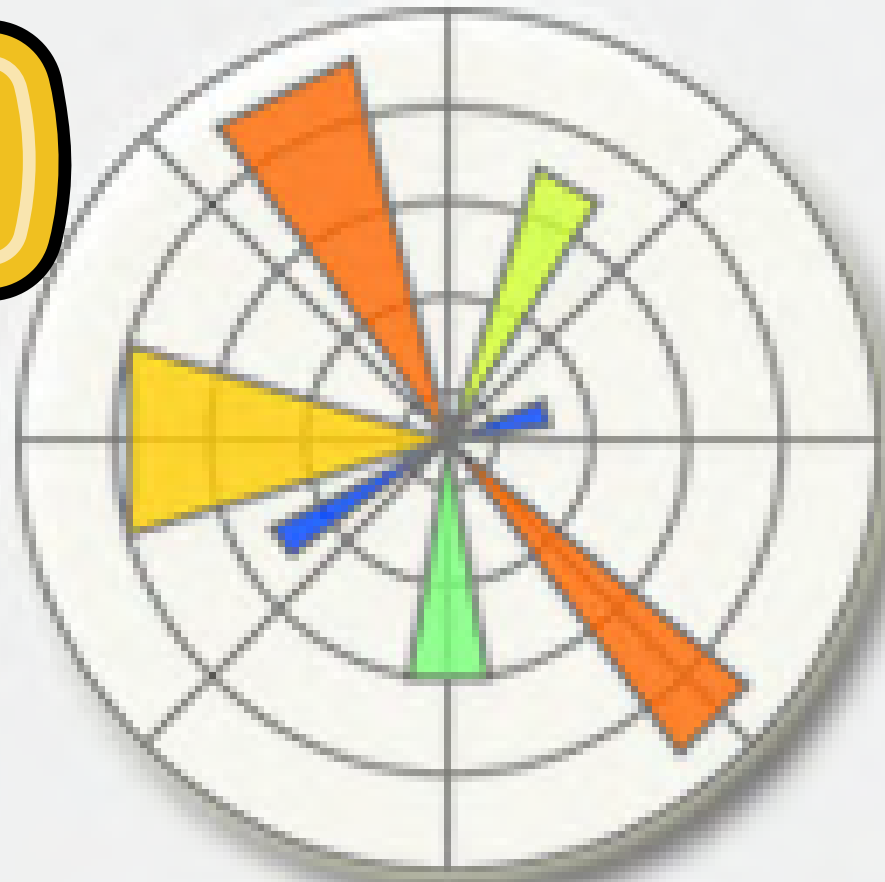
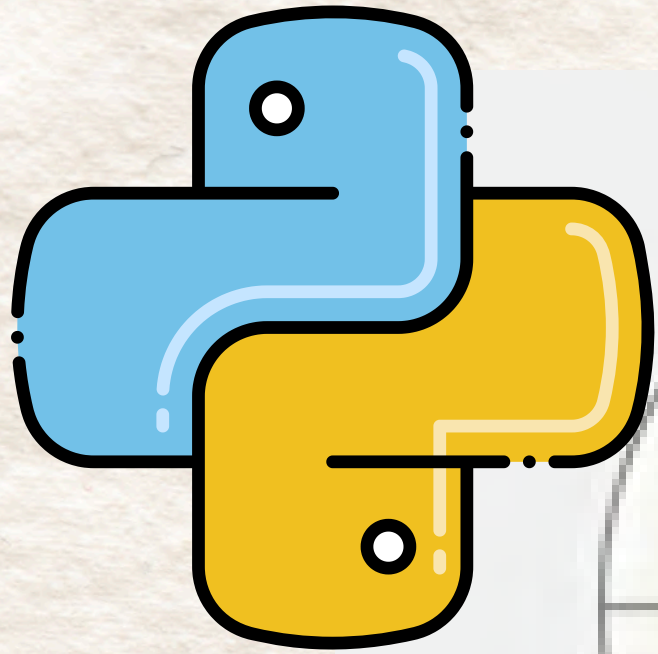
**3D charts
(distortion)**

Common Visualization Mistakes

**No title
or units**

**Misleading
axis labels**

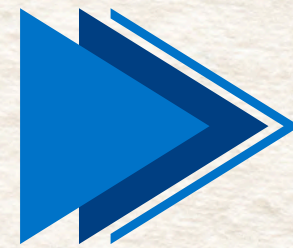
**Overcrowded
labels**



Matplotlib

Matplotlib Visualizations

What is Matplotlib?



Python plotting library



Highly customizable



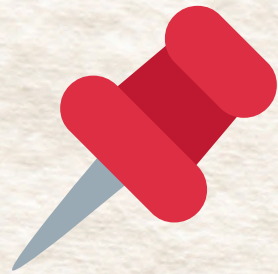
Base library for many others



Used for static visualizations

Bar Chart

(Category Comparison)



**Best for:
Showing
rankings**

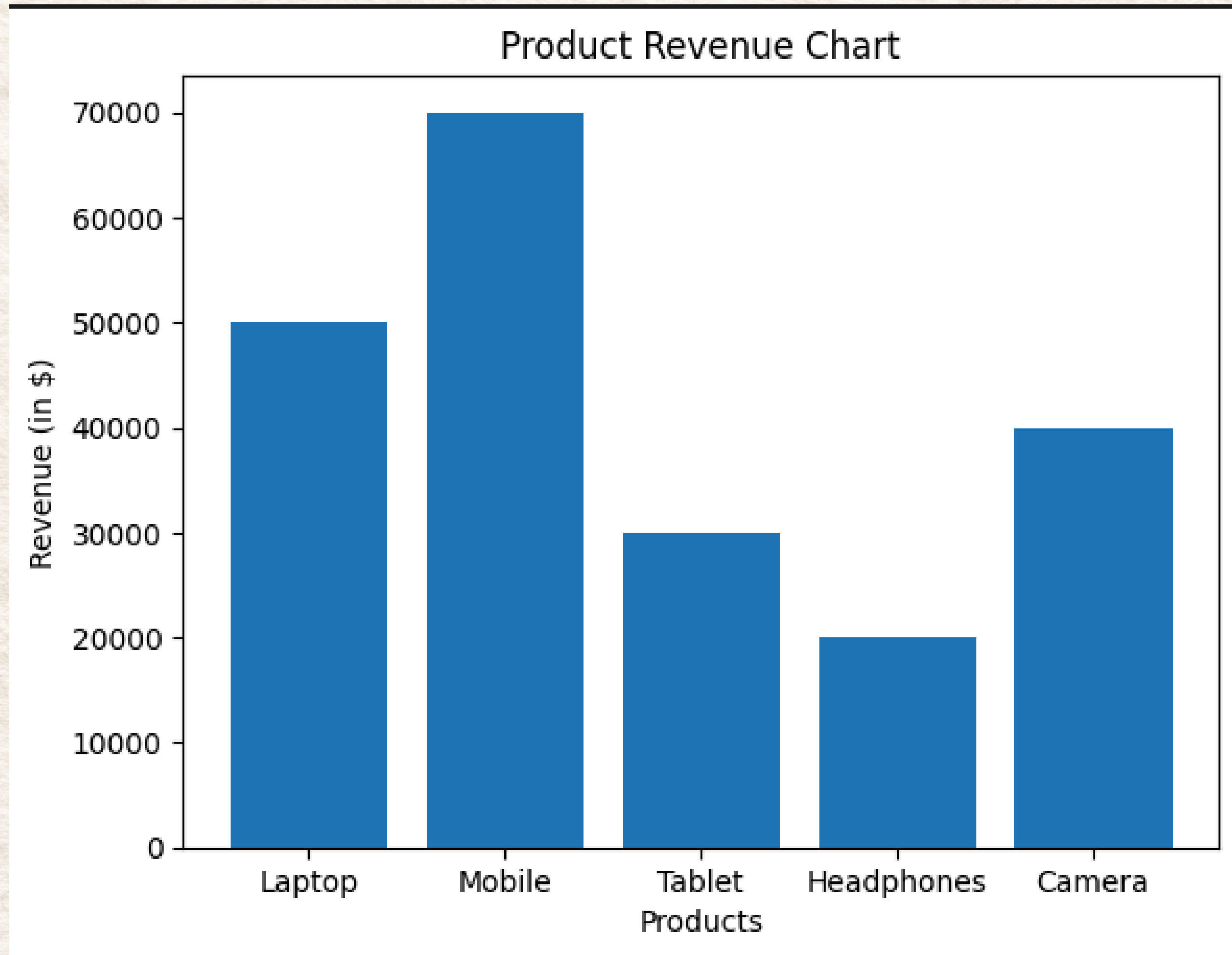
```
# Import the matplotlib library
import matplotlib.pyplot as plt

# Sample data
products = ['Laptop', 'Mobile', 'Tablet', 'Headphones', 'Camera']
revenue = [50000, 70000, 30000, 20000, 40000]

# Create bar chart
plt.bar(products, revenue)

# Add labels and title
plt.xlabel("Products")
plt.ylabel("Revenue (in $)")
plt.title("Product Revenue Chart")

# Display the chart
plt.show()
```

Scatter Plot (Relationship)

Use for:

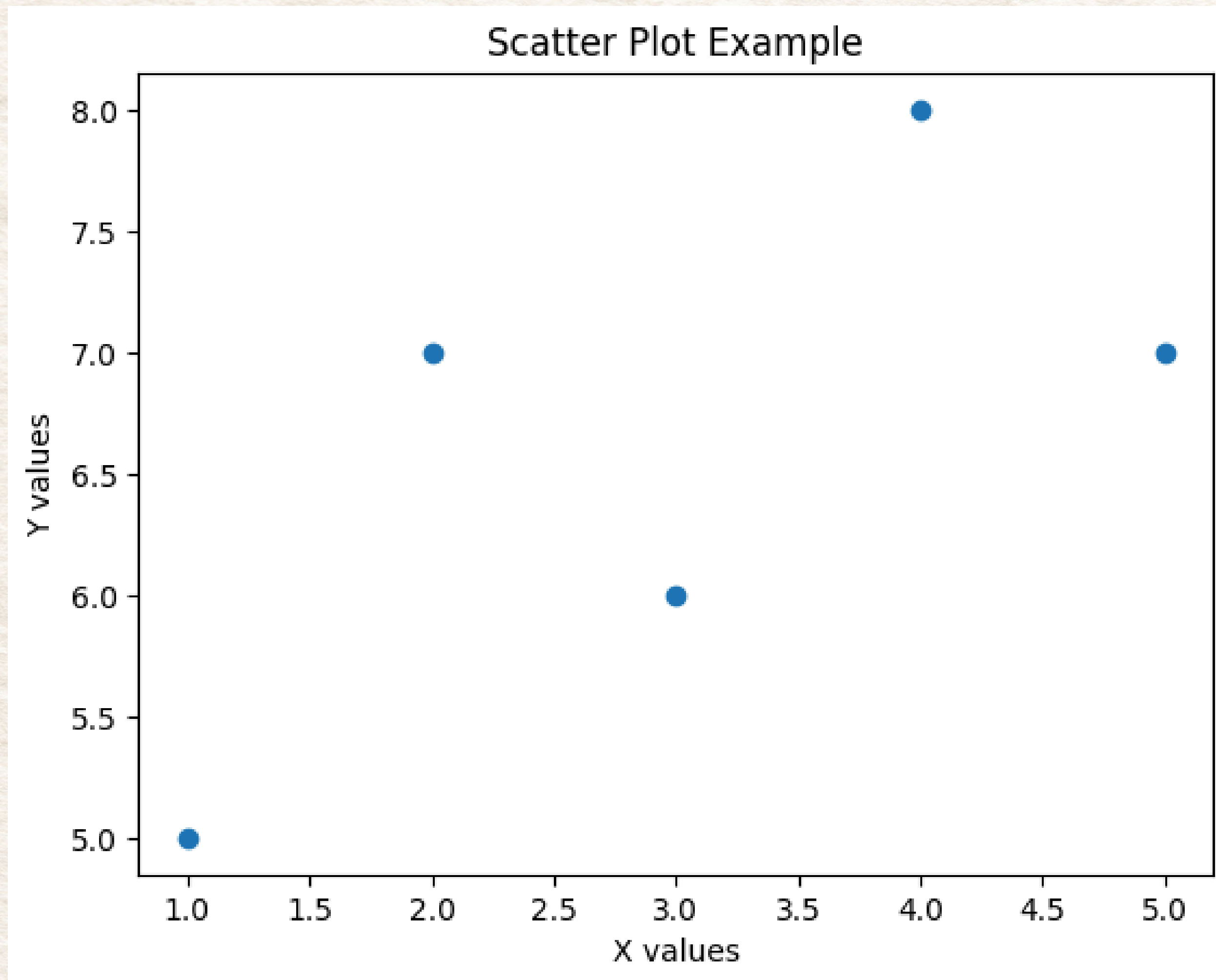
- **Correlation detection**
- **Trend pattern identification**

```
import matplotlib.pyplot as plt

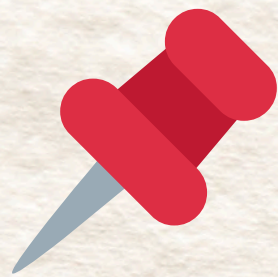
x = [1, 2, 3, 4, 5]
y = [5, 7, 6, 8, 7]

plt.scatter(x, y)
plt.title("Scatter Plot Example")
plt.xlabel("X values")
plt.ylabel("Y values")

plt.show()
```

Line Chart (Trend Analysis)



**Best for:
Time-series
analysis**

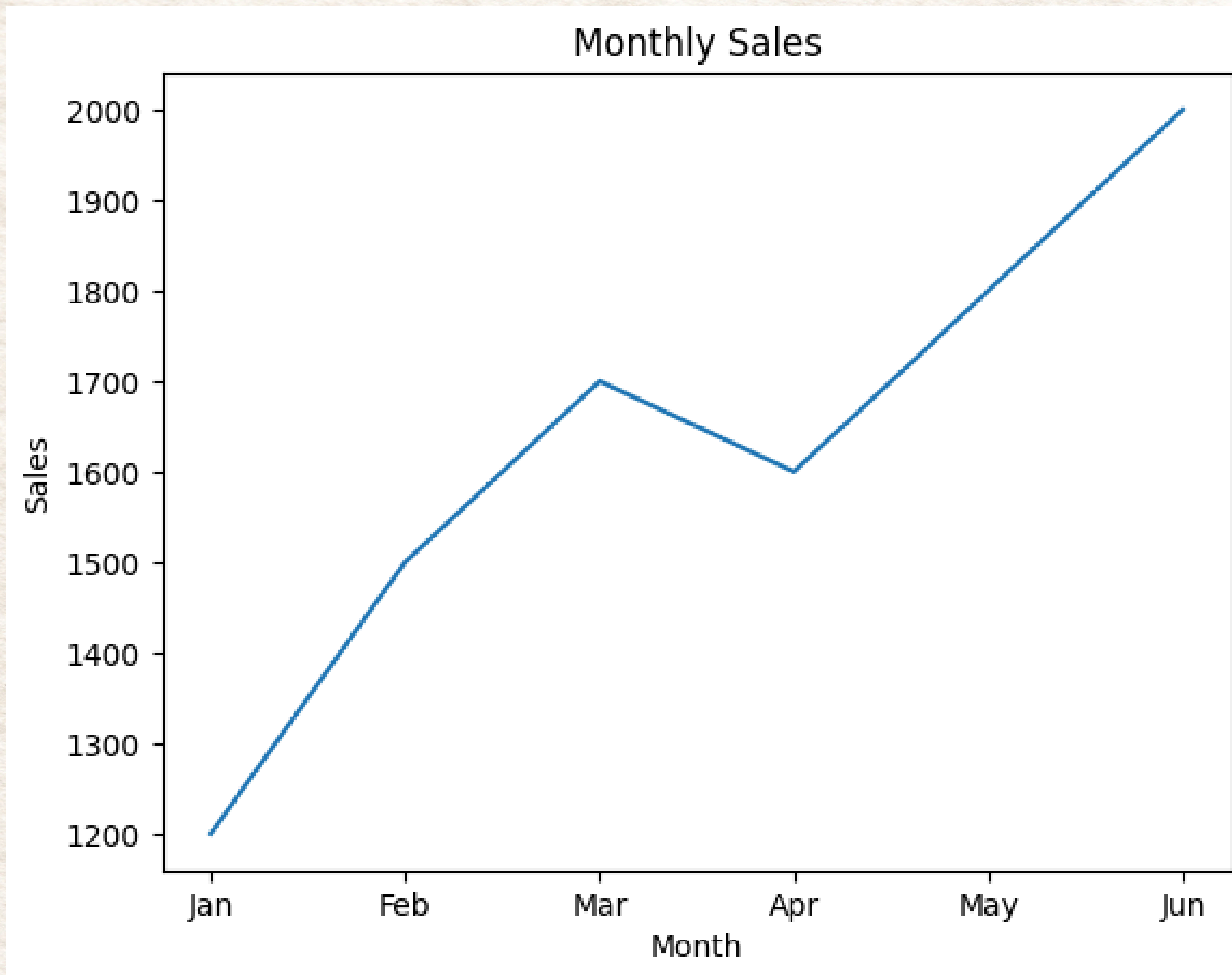
```
# Import matplotlib
import matplotlib.pyplot as plt

# Sample data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [1200, 1500, 1700, 1600, 1800, 2000]

# Create line plot
plt.plot(months, sales)

# Add title and labels
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

# Show the plot
plt.show()
```

Histogram (Distribution)

Shows:

- **Data spread**
- **Frequency**
- **Distribution pattern**

```
import matplotlib.pyplot as plt

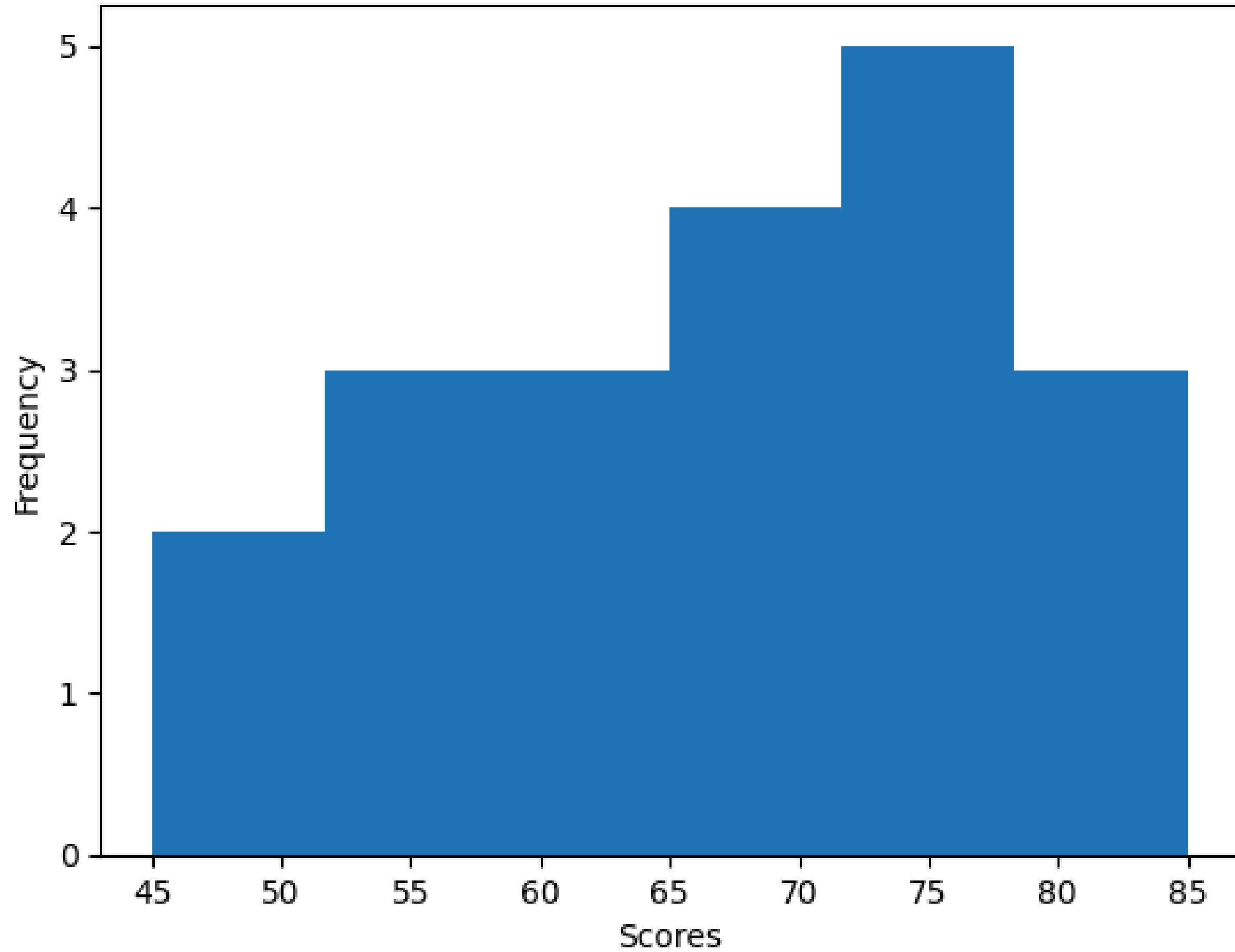
# Sample data
data = [45, 50, 52, 55, 57, 60, 62, 63, 65, 67,
        68, 70, 72, 74, 75, 77, 78, 80, 82, 85]

# Create histogram
plt.hist(data, bins=6)

# Labels and title
plt.title("Typical Histogram Example")
plt.xlabel("Scores")
plt.ylabel("Frequency")

plt.show()
```


Typical Histogram Example



Subplots

(Multiple Graphs Together)

Use when:

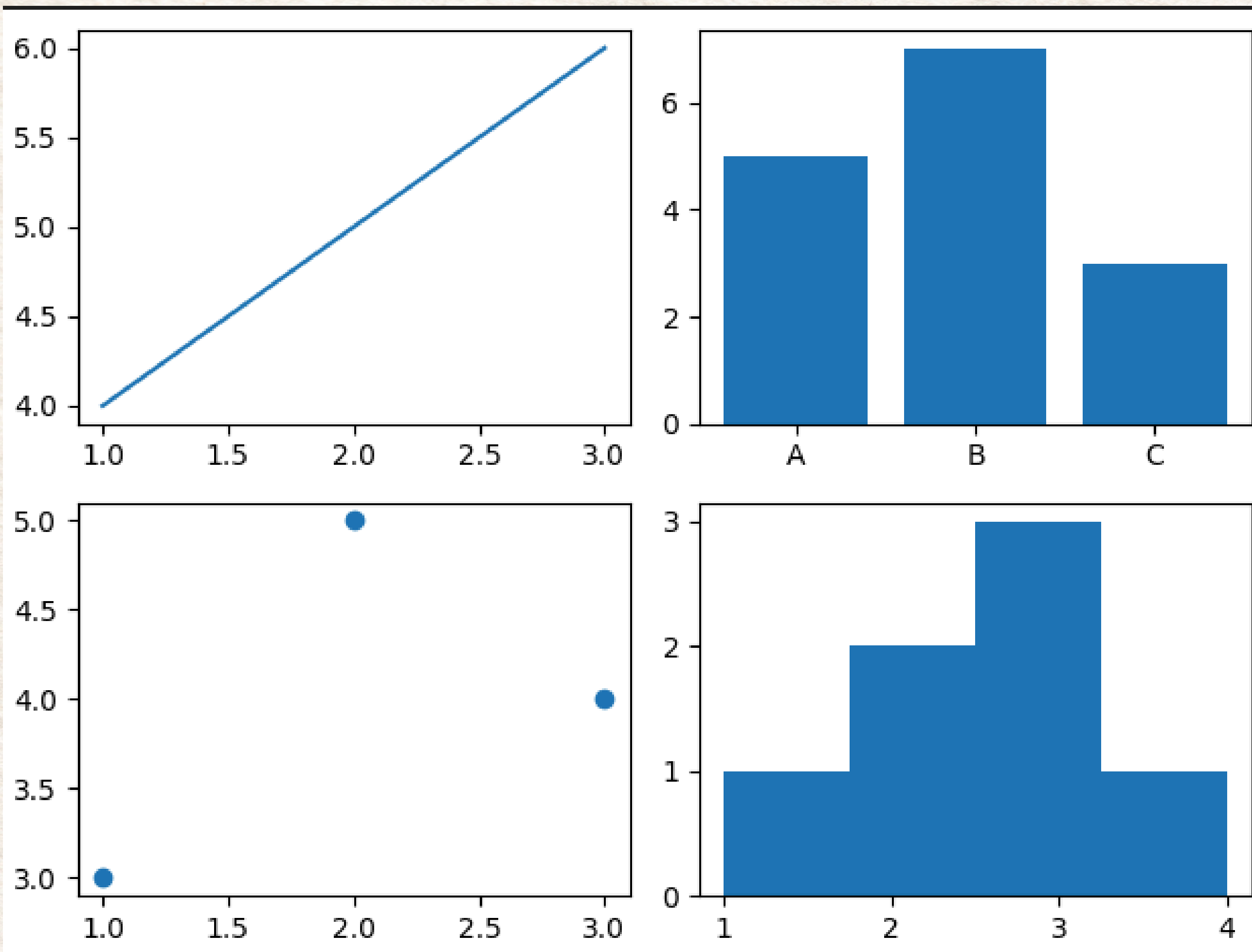
- Comparing multiple variables
- Creating mini dashboards

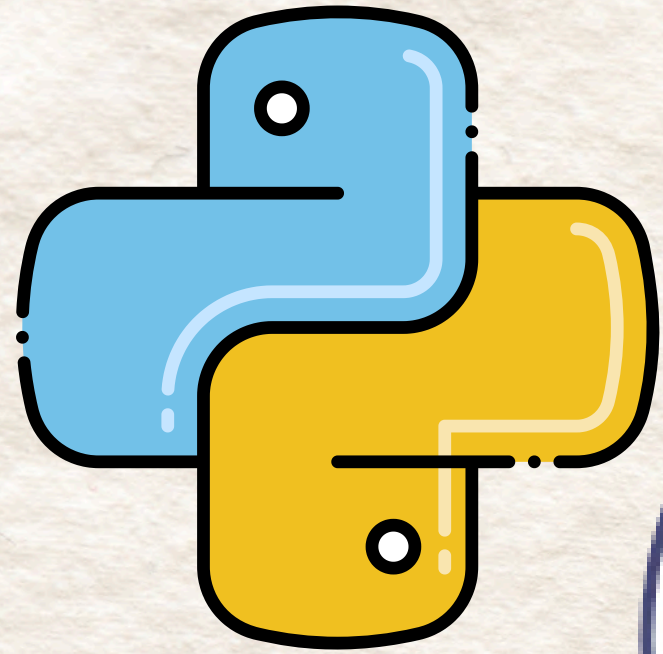
```
import matplotlib.pyplot as plt

fig, axs = plt.subplots(2, 2)

axs[0,0].plot([1,2,3], [4,5,6])
axs[0,1].bar(['A', 'B', 'C'], [5,7,3])
axs[1,0].scatter([1,2,3], [3,5,4])
axs[1,1].hist([1,2,2,3,3,3,4], bins=4)

plt.tight_layout()
plt.show()
```

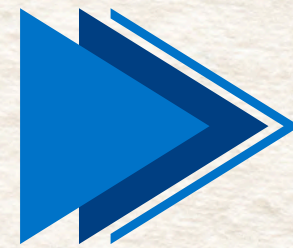





seaborn

Statistical Visualization with Seaborn

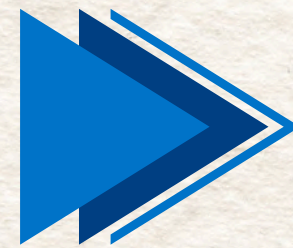
Why Seaborn?



Built on Matplotlib



Beautiful default styles



Statistical plottings



Less code, better visuals

Box Plot



Used for distribution comparison

Shows:

- **Median**
- **Quartiles**
- **Outliers**

A box plot (also called a box-and-whisker plot) is a type of graph used in statistics to show the distribution of numerical data.

It summarizes data using five important values and helps identify spread and outliers.


```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "gender": ["Male", "Female", "Male", "Female", "Male", "Female", "Male", "Female"],
    "salary": [50000, 52000, 55000, 60000, 58000, 62000, 54000, 61000]
}

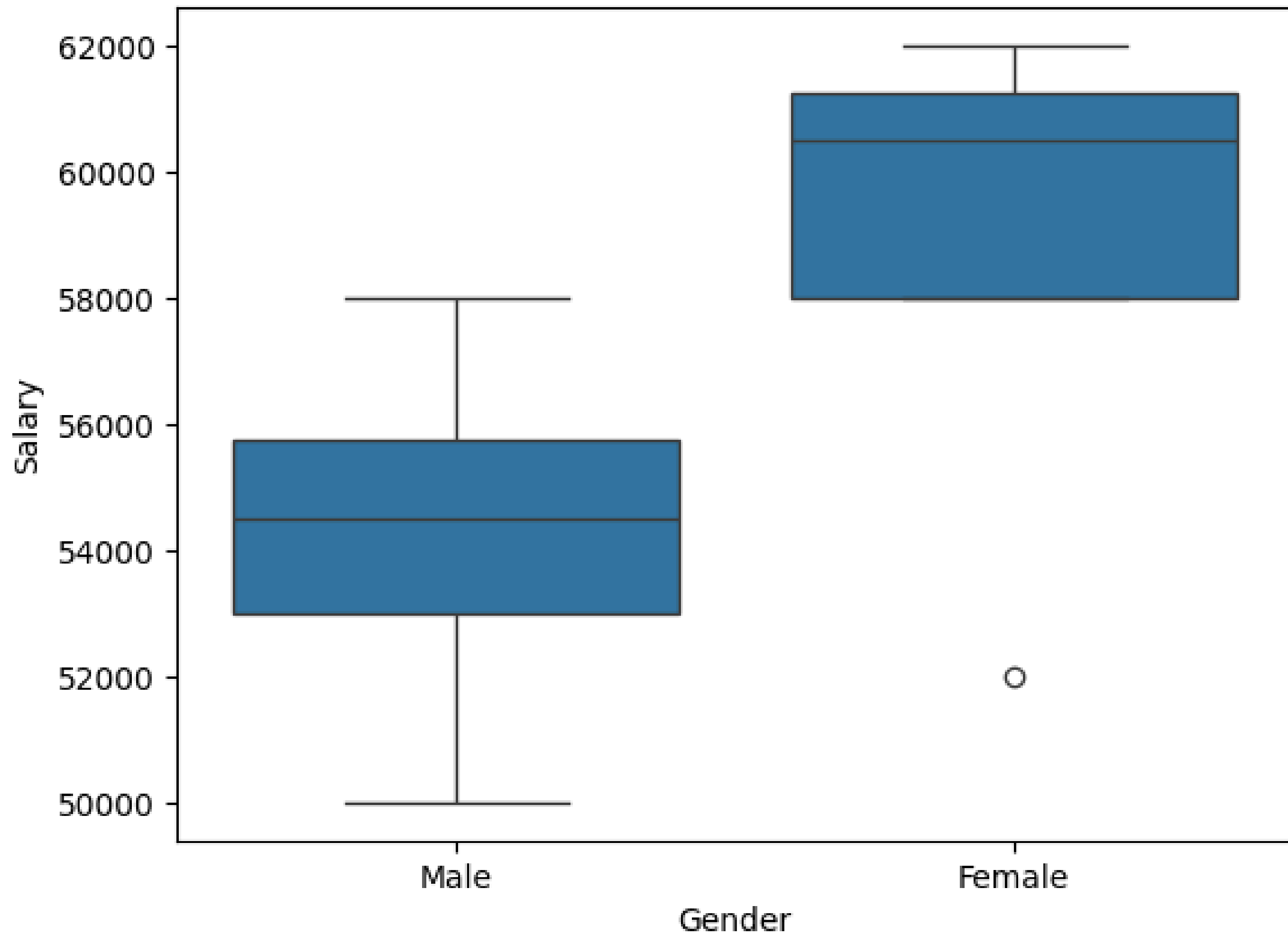
# Create DataFrame
df = pd.DataFrame(data)

# Create boxplot
sns.boxplot(x="gender", y="salary", data=df)

plt.title("Salary Distribution by Gender")
plt.xlabel("Gender")
plt.ylabel("Salary")

plt.show()
```


Salary Distribution by Gender



Pair Plot

Shows:

- Relationship between all numeric variables
- Distribution + scatter

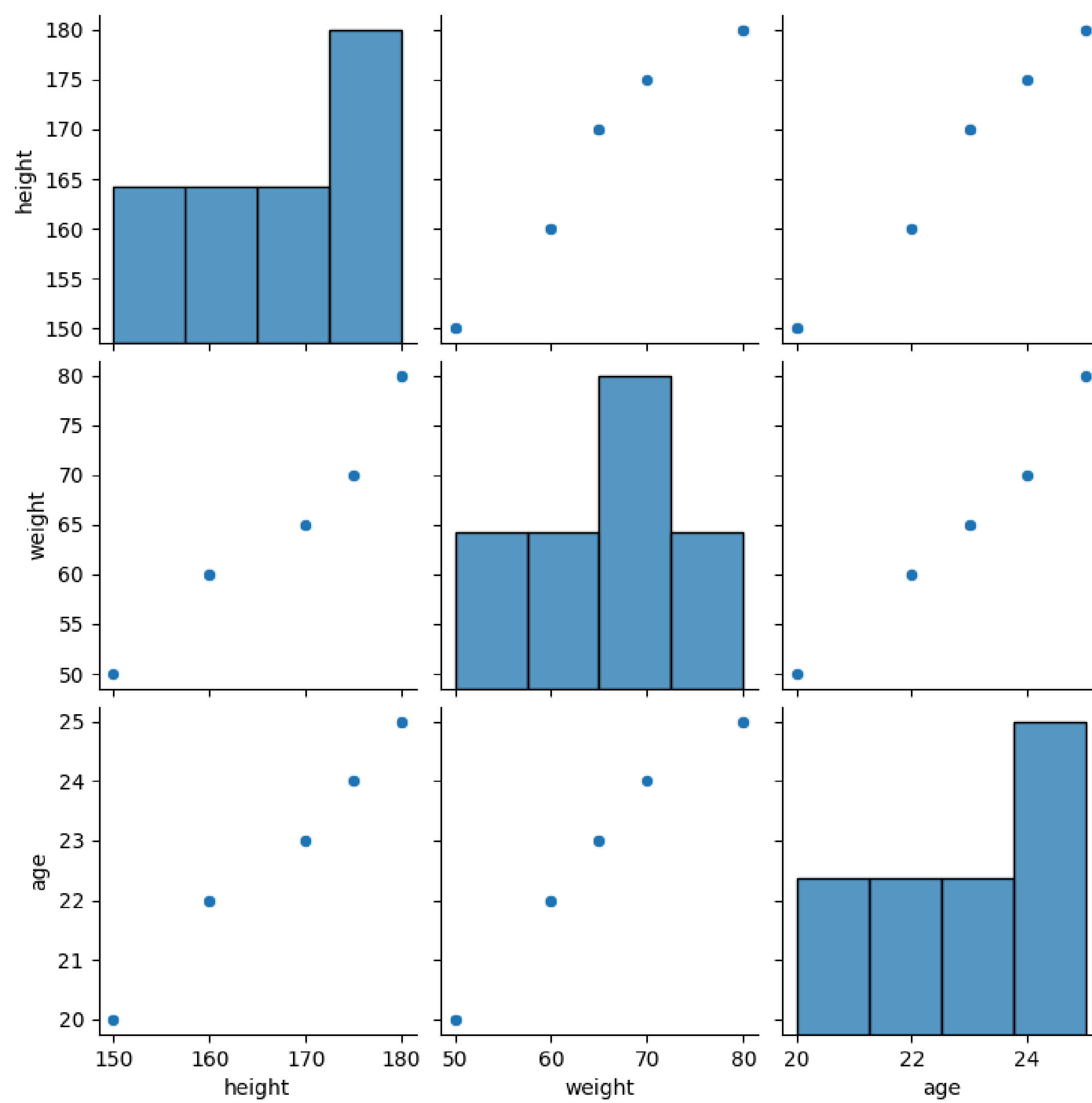
Useful for:

- Feature analysis
- Correlation detection

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "height": [150, 160, 170, 180, 175],
    "weight": [50, 60, 65, 80, 70],
    "age": [20, 22, 23, 25, 24]
}

df = pd.DataFrame(data)
sns.pairplot(df) #creating a pairplot
plt.show()
```

Heatmap (Correlation Matrix)

Shows:

- **Strength of correlation**
- **Positive / Negative relationship**

Dark color – Strong relationship

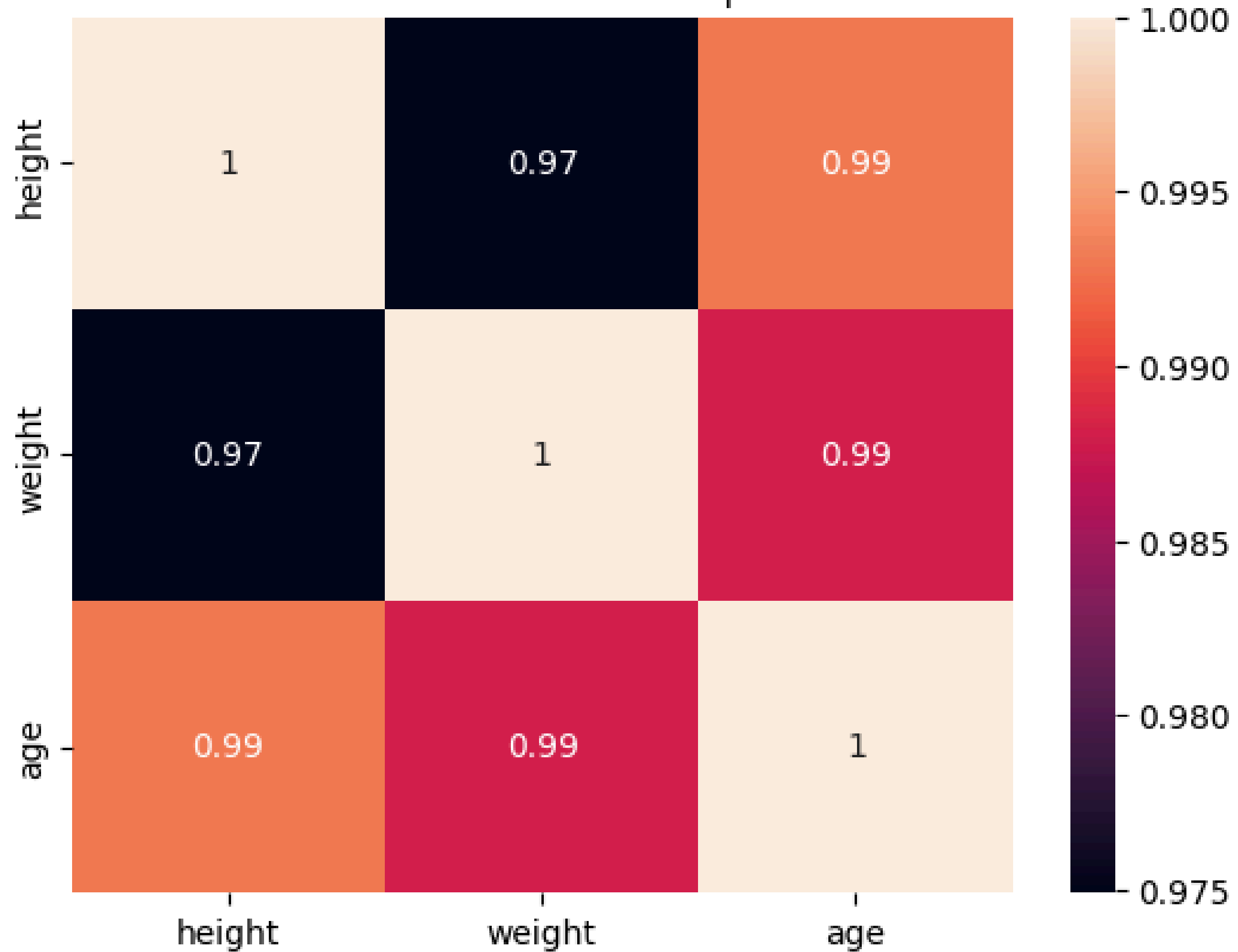
Light color – Weak relationship

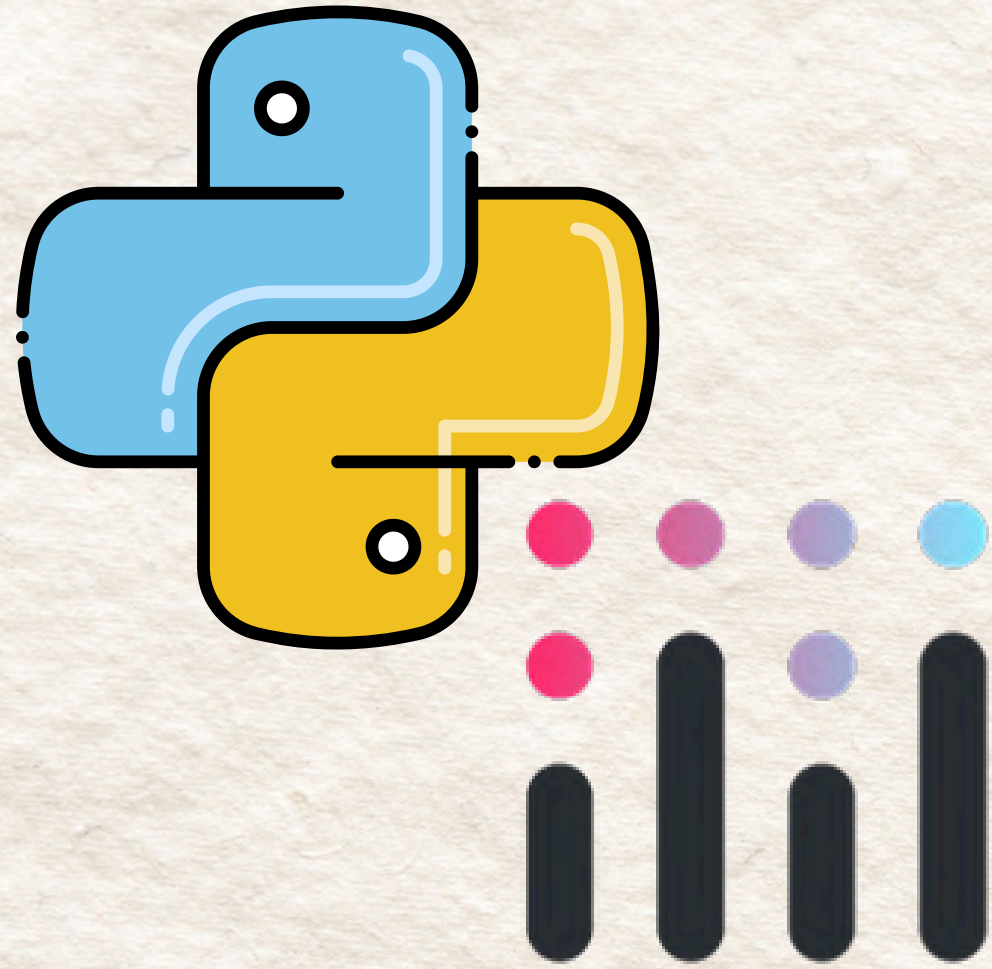
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
    "height": [150, 160, 170, 180, 175],
    "weight": [50, 60, 65, 80, 70],
    "age": [20, 22, 23, 25, 24]
}

df = pd.DataFrame(data)
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```


Correlation Heatmap





Interactive Visualization with Plotly

Why Interactive Visualization?

Static charts

Limited
Functionality

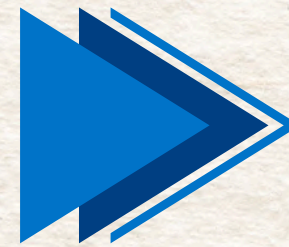
Interactive charts



Hover details



Zoom



Toggle categories



Filtering Data

Plotly

Features

- **Hover data**
- **Smooth UI**
- **Interactive legends**

Allows users to build interactive plots such as line charts, bar charts, scatter plots, and dashboards that can be zoomed, hovered, and explored.

Use Cases

- **Business dashboards**
- **Web apps**
- **Data storytelling reports**
- **Real-time analytics**

Plotly Example (Line Chart)



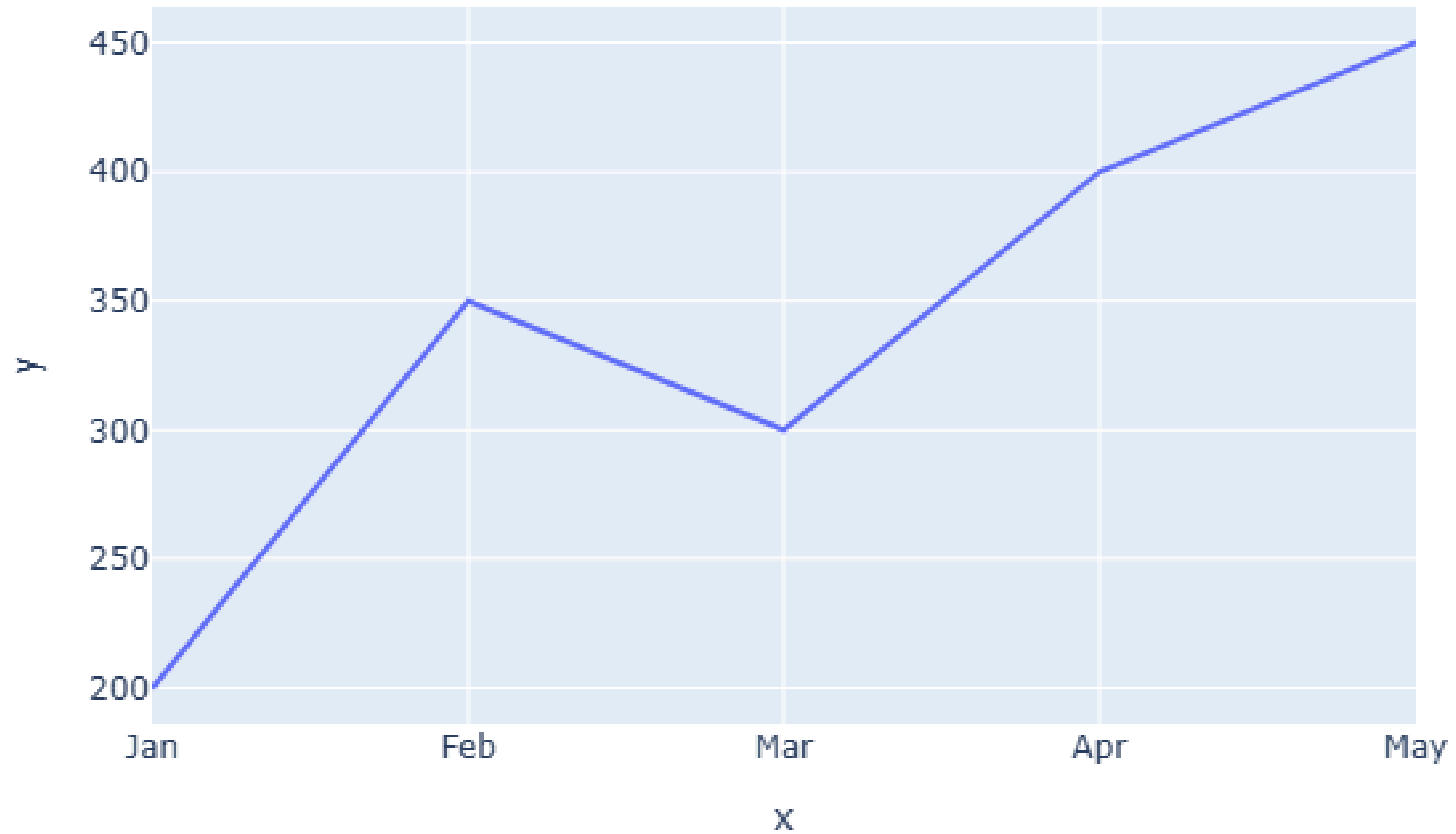
```
import plotly.express as px

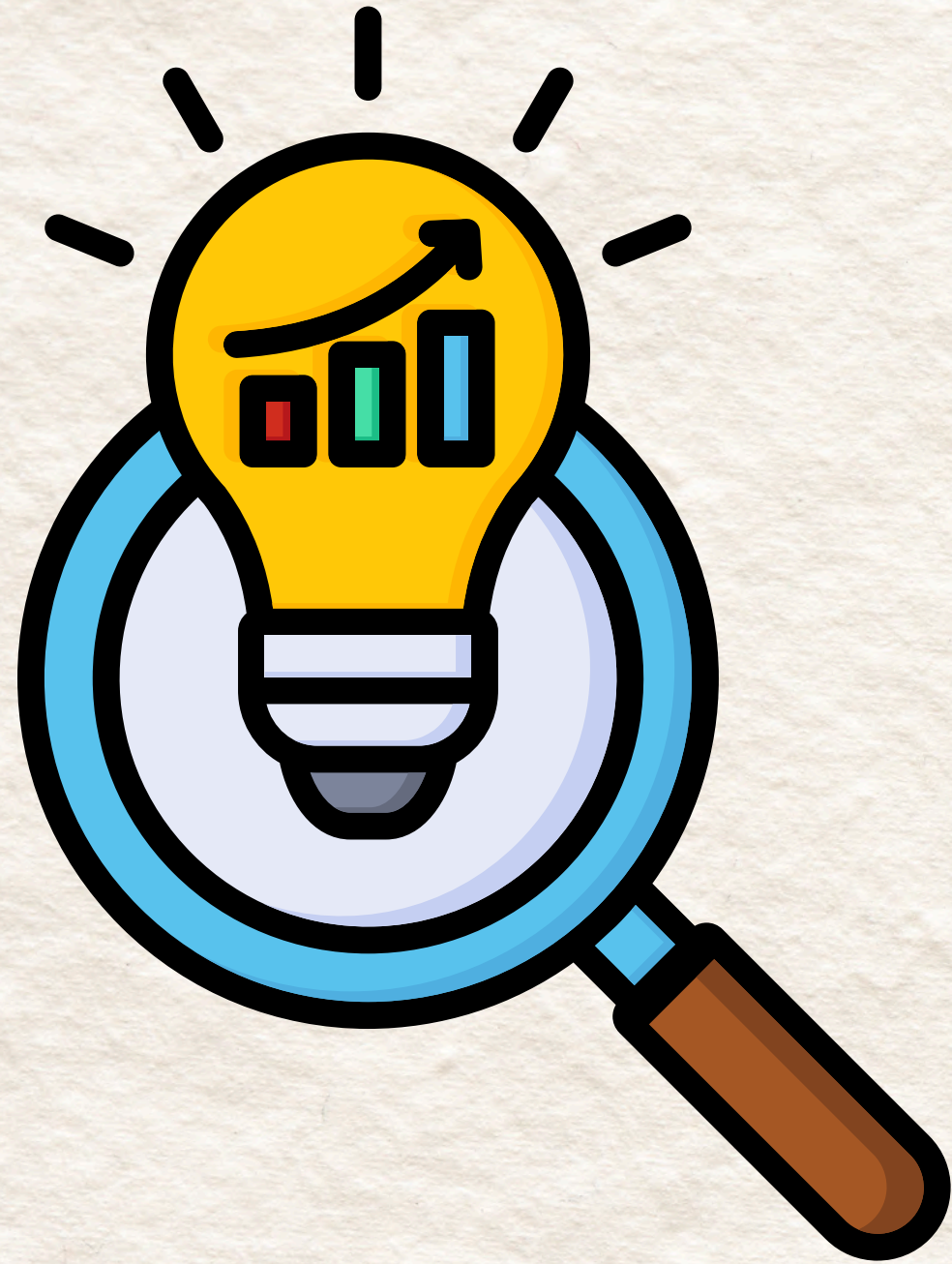
# Sample data
months = ["Jan", "Feb", "Mar", "Apr", "May"]
sales = [200, 350, 300, 400, 450]

# Create line plot
fig = px.line(x=months, y=sales, title="Monthly Sales")

fig.show()
```


Monthly Sales





Visualization-Driven Insight Generation

What is Insight?

Insight answers:

- **Why?**
- **What next?**
- **What should we change?**

Insight in Data Visualization means the meaningful understanding or conclusion obtained from a visual representation of data (charts, graphs, dashboards).

Example:

If a line chart of monthly sales shows sales rising from June to December, the insight is:

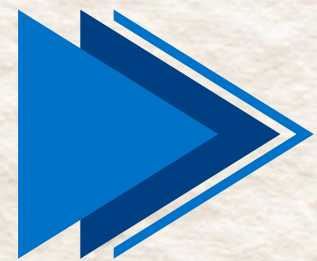
➡ Sales increase in the second half of the year.

Common insights from visualizations:

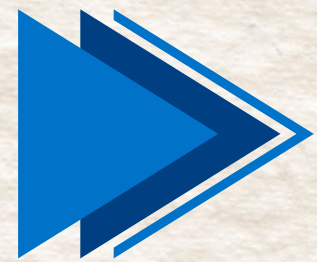
Data → Information → Insight → Decision



Trends – increase or decrease over time



Patterns – repeated behavior in data

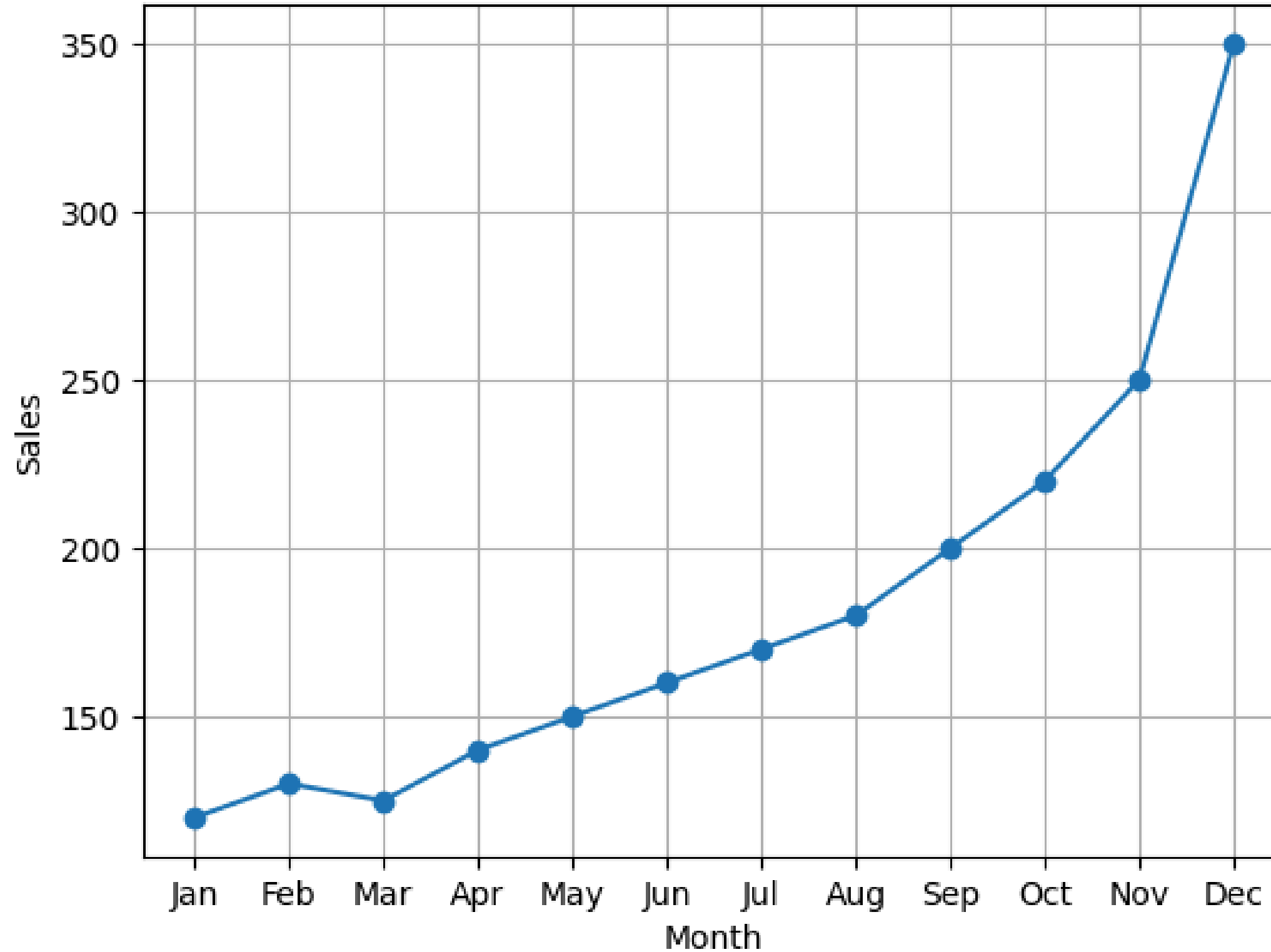


Relationships – how two variables are related



Outliers – unusual or extreme values

Monthly Sales Trend



Observation:

Sales increase sharply in December



Insight:

Holiday demand boost



Decision:

Increase inventory before December

Insight Framework

For insight



Observe pattern



Ask why



Validate with data



Recommend action



Case Study: End-to-End Visualization Project

Retail Sales Analysis Project

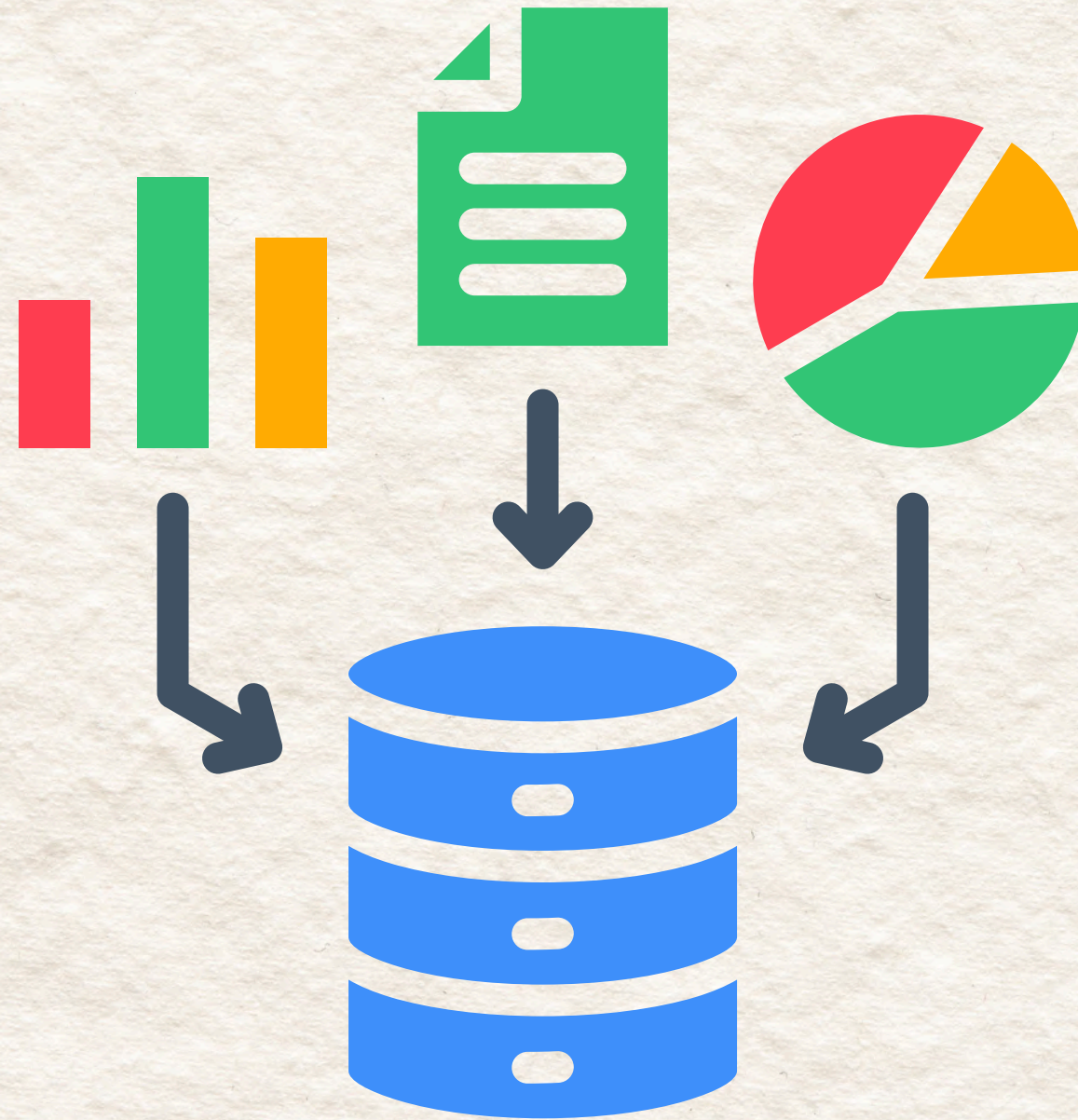


Goals:

- Understand sales trend
- Identify top products
- Analyze customer behavior
- Improve profit

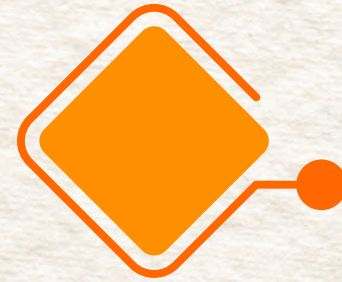
Step 1 – Data Collection

Data Sources:
CSV file
Database
APIs

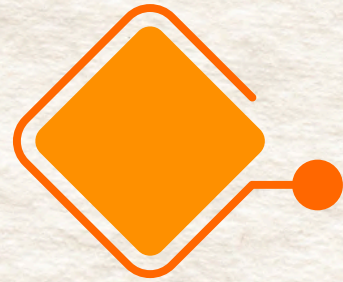


Example fields:
Date
Product
Revenue
Customer Age
Region

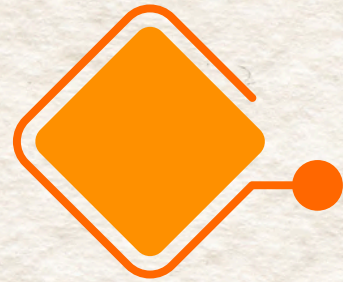
Step 2 – Data Cleaning



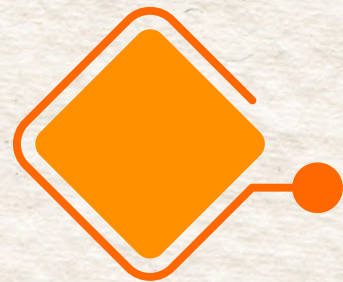
Handle missing values



Format dates



Normalize categories



Remove duplicates

Step 3

Exploratory Visualization

-  **Line chart → Monthly sales**
-  **Bar chart → Top products**
-  **Histogram → Age distribution**
-  **Heatmap → Correlation**

Step 4 – Insight Generation



Findings:

- **Sales peak in December**
- **Young adults buy more electronics**
- **High correlation between advertising and revenue**

Step 5 – Dashboard Creation

Combine:

- KPI cards
- Trend graph
- Category comparison
- Region analysis



Tools:
Plotly
Power BI
Tableau

Final Report Structure

The final report of a case study should follow the format

- 1. Introduction**
- 2. Data description**
- 3. Key visualizations**
- 4. Insights**
- 5. Recommendations**
- 6. Conclusion**



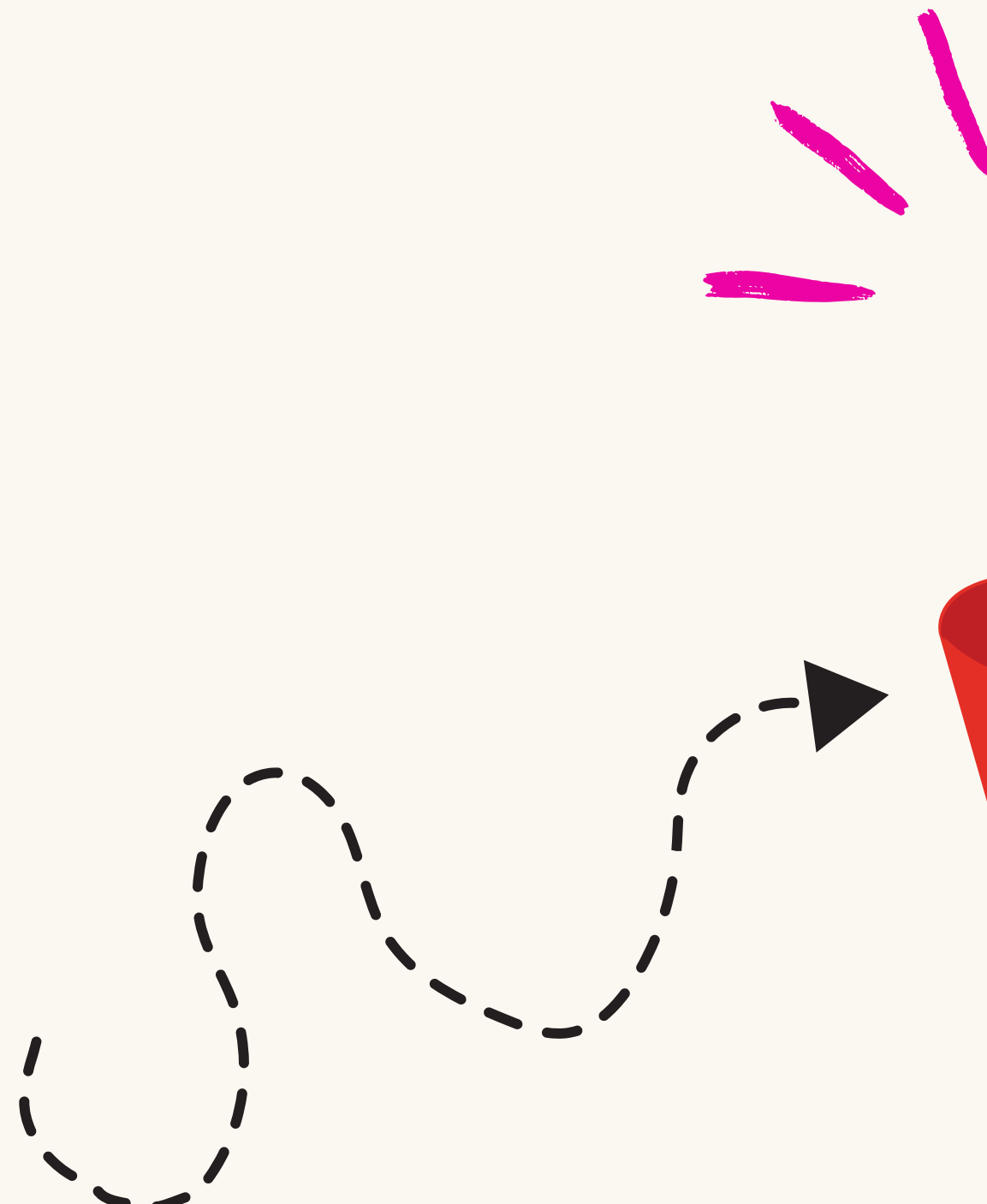


Conclusion

**Data visualization is not about charts.
It is about telling stories that drive action.**

**"Without data, you're just another person
with an opinion."**

— W. Edwards Deming



Thank You!

